

Q.25 Write short note on any three :-

- 1) IUPAC Naming of alkane
- 2) List of petrochemical plant in India
- 3) Significance of the petrochemical industry in India's economy
- 4) Classify petroleum

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Roll No.

6th Sem/ Chemical (Pulp & Paper)
Subject : Petroleum Technology & Petrochemical

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is the main component of petroleum?
- a) Methane b) Ethylene
c) Hydrocarbons d) Carbon dioxide
- Q.2 Vinyl chloride is used as a raw material to produce:
- a) Nylon
b) PVC (Polyvinyl chloride)
c) Teflon
d) PET
- Q.3 The IUPAC name of $\text{CH}_3\text{-CH}_2\text{-OH}$ is :
- a) Methanol b) Ethanol
c) Propanol d) Butanol
- Q.4 Which of the following safety equipment is used to prevent chemical inhalation?
- a) Fire extinguisher b) Gas mask
c) Gloves d) Goggles

- Q.5 Which fire extinguisher is used for electrical fires?
a) Water b) Foam
c) Carbon dioxide d) Wet chemical
- Q.6 Which functional group is present in acetone?
a) Alcohol b) Aldehyde
c) Ketone d) Ester

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Petrochemicals are derived from.
- Q.8 What is fractional distillation.
- Q.9 _____ is commonly used as a solvent in laboratories and industries.
- Q.10 The most common fire extinguisher for electrical fires is _____.
- Q.11 Writes the chemical formula for propane.
- Q.12 The functional group - OH represents _____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 List the various fractions obtained, their boiling ranges, and uses with diagram.
- Q.14 Describe the manufacturing process of phenol with a neat flow diagram.

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- Q.15 Explain different types of fire and explosion hazards in the petroleum and chemical industries.
- Q.16 Describe the IUPAC naming of cyclic, aromatic and functional group compounds with at least two examples for each.
- Q.17 Explain the manufacturing process of methanol with a flow sheet.
- Q.18 Define petrochemicals. What are the main raw materials used in the petrochemical industry?
- Q.19 Writes down about production of butanol.
- Q.20 Discuss the methods used for the purification of petroleum products.
- Q.21 What are the major uses of C3 and C4 compounds in the chemical industry?
- Q.22 What are the common hazards and how can they be prevented?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the manufacturing process of acetone in detail. Provide a flow sheet and also describe the engineering problems involved.
- Q.24 Discuss about the process of phenol, including the chemical reactions involved, raw material and process conditions with flow sheet.

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